

LIÈGE université
Sciences Appliquées

Transient Stability

ELEC0447 - Analysis of electric power and energy systems

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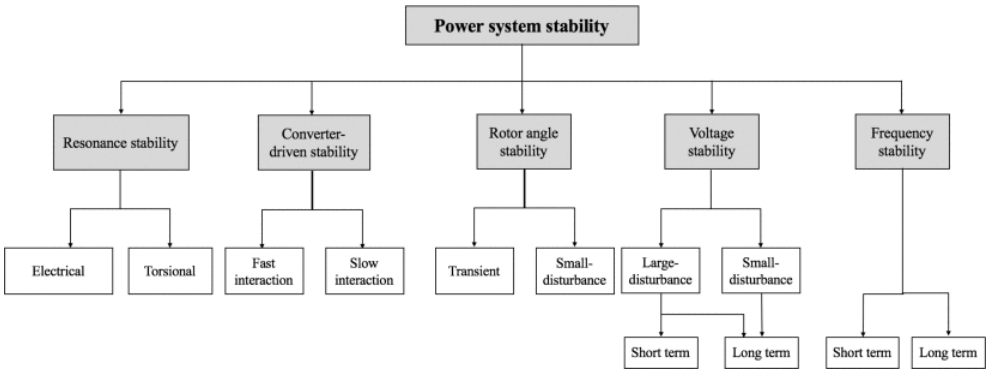
Overview

1. Principle of transient stability
2. The Equal Area Criterion
3. Practical Example
4. Short-Circuit Capacity and Rotor Angle Stability
5. Dynamic Security Assessment
6. Trends

Classification power system stability (Reminder) I

Power system stability is the ability of an electric power system, for a given initial operating condition, to regain a state of operating equilibrium after being subjected to a physical disturbance, with most system variables bounded so that practically the entire system remains intact. *[1]*

Classification power system stability (Reminder) II



Principle of transient stability

Reminder: generator model and step up transformer I

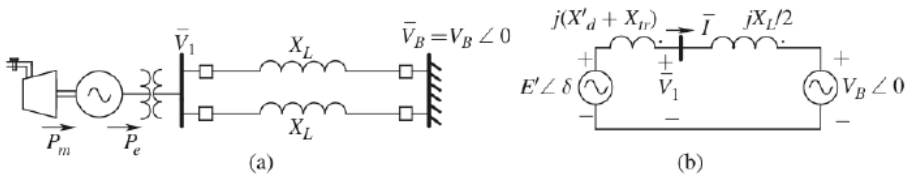


Figure 1: Generator with step-up transformer connected to infinite bus through two parallel lines.

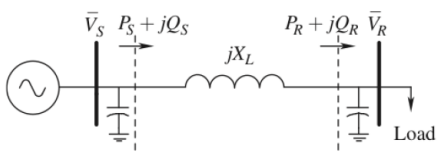
- Consider a generator bus connected through a transformer to an infinite bus system ($\bar{V}_B = V_B$ is an ideal voltage source).

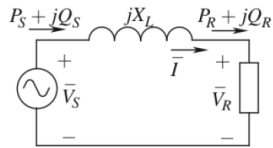
Reminder: generator model and step up transformer II

- The mechanical power delivered through the shaft is P_m.
- The electrical power delivered to the grid is denoted P_e.
- In steady-state, the synchronous machine is modelled as a constant voltage source $E'e^{j\delta}$ behind its transient reactance X'_d.
- Finally, X_{tr} is the leakage reactance of the transformer.

Reminder: active and reactive power flow on a line I

Consider a simple radial system.





Reminder: active and reactive power flow on a line II

Assuming no transmission-line losses:

$$S_S = P_S + jQ_S = V_S e^{j\delta_S} \left(\frac{V_S e^{-j\delta_S} - V_R e^{-j\delta_R}}{X} \right) e^{j\frac{\pi}{2}}$$

If we define $\delta = \delta_S - \delta_R$, we have:

$$P_R = P_S = \frac{V_S V_R}{X_L} \sin \delta$$

$$Q_R = \frac{V_S V_R \cos \delta}{X_L} - \frac{V_R^2}{X_L}$$

$$Q_S = \frac{V_S^2}{X_L} - \frac{V_S V_R \cos \delta}{X_L}$$

Reminder: Electrical power vs. angle

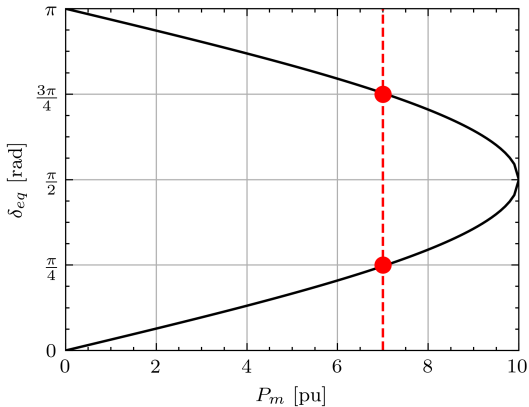
Plugging the generator and transformer models in the previous models, the electrical power P_e is expressed as follows:

$$P_e = \frac{E' V_B}{X_{tr} + X'_d + X_L/2} \sin(\delta) \tag{1}$$

At equilibrium, $P_e = P_m$. For given reactances and voltage sources, one has:

$$\delta_{eq} = \arcsin\left(P_m \frac{X_{tr} + X'_d + X_L/2}{E' V_B} \right)$$

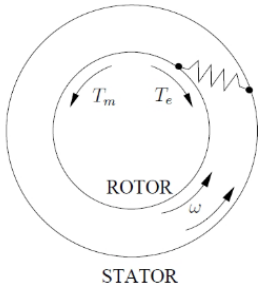
Reminder: Electrical power vs. angle, graphically



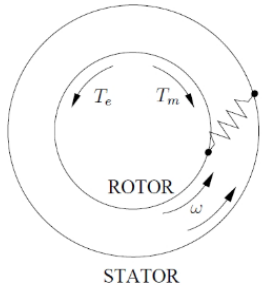
- When we increase P_m, the electrical angle at equilibrium also increases.
- The maximum value is reached for $\delta_{eq} = \frac{\pi}{2}$.
- There exist
  - **TWO** equilibrium points for $P_m \in [0, 10)$
  - **ONE** for $P_m = 10$
  - **NONE** for $P_m > 10$.

Reminder: what if P_m exceeds P_e? I

(In this case, $P_m > 10$.)



generator operation



motor operation

Reminder: what if P_m exceeds P_e? II

- In synchronous machines, the rotor rotates at the same angular speed as the magnetic field produced by the stator.
- If we denote ω_m the rotor angular speed and $\omega_s = 2\pi f_s$ the synchronous speed, then $\omega_m = \omega_s$.
- The field produced by the rotor and the stator tend to align.
- If they are not aligned, an electromechanical torque is produced.
- This misalignment is represented by the electrical angle: a greater angle induces a larger torque.
- **BUT** if the angle is too large, the machine loses synchronism and the torque becomes 0.

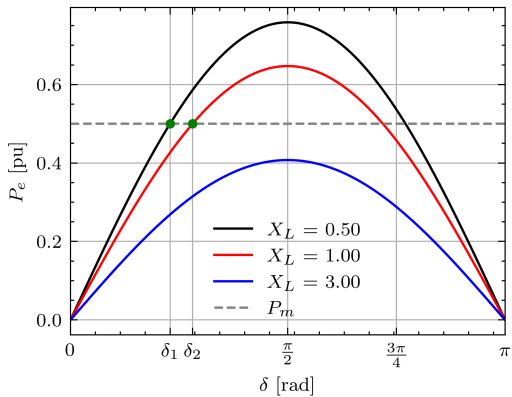
Principle of transient stability

Consider equation (1) again:

$$P_e = \frac{E' V_B}{X_{tr} + X'_d + X_L/2} \sin(\delta)$$

with (all in per unit)

- $E' = 1.1, V_B = 1, X_{tr} = 0.8, X'_d = 0.4$
- various values of $X_L = [0.5, 1, 3]$ (see next slide).



Let us consider the following sequence of actions

Refer to Figure 1:

1. **Pre-fault conditions**: the impedance between the generator and the infinite bus system is $jX_L = j0.5$ pu.
2. **During-fault conditions**: one line is short-circuited, the impedance becomes $jX_L = j3$ pu.
3. **Post-fault conditions**: the short-circuit is cleared by disconnecting a line, the impedance becomes $jX_L = j1$ pu.

For the system to be stable, the system has to stabilize at $\delta = \delta_2$ after the fault occured. *Why?*

We will now look at how the system moves from the equilibrium point δ_1 to equilibrium point δ_2

The Equal Area Criterion

The swing equation (reminder) I

Let us define the following quantities: J_m is the moment-of-inertia of the rotational system, T_m is the mechanical torque, T_e is the electrical torque, ω_m is the rotor speed and δ_m the rotor angle (in mechanical radians). Using Newton's Second Law, one has (we omit the damping torque for now, we will discuss its impact later.):

$$J_m \frac{d^2\delta_m}{dt^2} = T_m - T_e$$

Multiplying both sides by ω_m (the rotor speed), one has:

$$\omega_m J_m \frac{d^2\delta_m}{dt^2} = P_m - P_e$$

The swing equation (reminder) II

We then define

$$H_{gen} = \frac{\frac{1}{2} J_m \omega_{syn,m}^2}{S_{rated,gen}} = \left[\frac{kg \cdot m^2 \times 1/s^2}{kg \cdot m^2/s^3} \right] = [s]$$

where $\omega_{syn,m}$ is the synchronous speed (in mechanical radians/s), $S_{rated,gen}$ the nominal rated size of the generator.

The swing equation (reminder) III

About the inertia constant H_{gen}

The inertia constant of a machine H_{gen} is a value that quantifies how much kinetic energy is stored in the rotating mass of a machine relative to its power rating. It indicates the duration for which a machine can supply its rated power using only its stored rotational kinetic energy when the input power is suddenly removed. It takes a value in a range of

- 3 to 11s for turbo-alternators
- 1 to 2s for hydro generators.

The swing equation (reminder) IV

One can write the swing equation by substituting J_m with H_{gen}:

$$\left(\frac{\omega_m}{\omega_{syn,m}^2}\right) 2H_{gen}\frac{d^2\delta_m}{dt^2} = P_{m,gen,pu} - P_{e,gen,pu}$$

where P_m and P_e are in per-unit of the generator MVA base. One can also write the equation in the system base S_{system}:

$$\left(\frac{\omega_m}{\omega_{syn,m}^2}\right) 2H\frac{d^2\delta_m}{dt^2} = P_{m,pu} - P_{e,pu}$$

The swing equation (reminder) V

with

$$H = H_{gen}\left(\frac{S_{rated,gen}}{S_{system}}\right)$$

Finally, we make the reasonable assumption that $\omega_m \approx \omega_{syn,m}$, and we express everything in terms of electrical radians.

$$\left(\frac{2H}{\omega_{syn}}\right)\frac{d^2\delta}{dt^2} = P_{m,pu} - P_{e,pu} \tag{2}$$

What can we say about the swing equation?

- If the mechanical power provided at the shaft P_m is greater than the electrical power transferred to the network P_e, the machine accelerates $\frac{d^2\delta}{dt^2} = \frac{d\Delta\omega}{dt} > 0$, where $\Delta\omega$ is the deviation from the synchronous speed ω_{syn}.
- It decelerates if the electrical power is greater than the mechanical power.
- The acceleration is proportional to the machine inertia H

Question

What is the inertia of PV panels?

What is a criterion for transient stability? I

Criterion

Assume δ_0 is an equilibrium point and an incident occurs. For the system to stabilize, it must exist an angle δ_m at which the kinetic energy becomes 0, and thus:

$$\int_{\delta_0}^{\delta_m} (P_{m,pu} - P_{e,pu})\, d\theta = 0$$

Demonstration: The swing equation established in (2) is

$$\left(\frac{2H}{\omega_{syn}}\right) \frac{d^2\delta}{dt^2} = P_{m,pu} - P_{e,pu}$$

What is a criterion for transient stability? II

Let's rearrange the terms and multiply both sides by $d\delta/dt$:

$$2\frac{d\delta}{dt}\frac{d^2\delta}{dt^2} = \frac{\omega_{syn}}{H}\left(P_{m,pu} - P_{e,pu}\right)\frac{d\delta}{dt}$$

Applying change of variables $\delta = \theta$, and integrating both sides between δ_0 and an arbitrary angle δ:

$$\int_{\delta_0}^{\delta}\left(2\frac{d\theta}{dt}\frac{d^2\theta}{dt^2}\right)dt = \frac{\omega_{syn}}{H}\int_{\delta_0}^{\delta}\left(P_{m,pu} - P_{e,pu}\right)d\theta$$

For the left hand side, we observe that the integrand is of the form $2f(t)f'(t)$, which suggests a **u-substitution**. Let the function be $u = \frac{d\theta}{dt}$.

What is a criterion for transient stability? III

The differential of u with respect to t is:

$$\frac{du}{dt} = \frac{d}{dt}\left(\frac{d\theta}{dt}\right) = \frac{d^2\theta}{dt^2}$$

This implies that $du = \frac{d^2\theta}{dt^2}dt$. By substituting u and du into the original integral, we transform it into a simpler form:

$$\int 2u\, du$$

The indefinite integral of this expression is:

$$\int 2u\, du = 2\left(\frac{u^2}{2}\right) + C = u^2 + C$$

What is a criterion for transient stability? IV

Now, we substitute back the original expression for u:

$$u^2 = \left(\frac{d\theta}{dt}\right)^2$$

Finally, we apply the limits of integration from $t = \delta_0$ to $t = \delta$ for the definite integral:

$$\int_{\delta_0}^{\delta} \left(2\frac{d\theta}{dt}\frac{d^2\theta}{dt^2}\right) dt = \left[\left(\frac{d\theta}{dt}\right)^2\right]_{\delta_0}^{\delta}$$

Evaluating the expression at the upper and lower limits:

$$= \left(\frac{d\theta}{dt}\right)^2\Bigg|_{t=\delta} - \left(\frac{d\theta}{dt}\right)^2\Bigg|_{t=\delta_0}$$

What is a criterion for transient stability? V

Since δ_0 is an equilibrium point, *i.e.* $d\delta/dt|_{\delta=\delta_0} = 0$, we have:

$$\left(\frac{d\delta}{dt}\right)^2 = \frac{\omega_{syn}}{H} \int_{\delta_0}^{\delta} (P_{m,pu} - P_{e,pu})\, d\theta$$

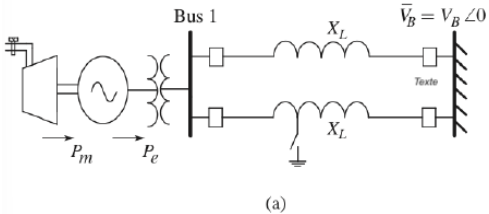
The left term $\left(\frac{d\delta}{dt}\right)^2 = \Delta\omega^2$ represents the kinetic energy of the machine at an arbitrary angle δ^* (with respect to the synchronous speed ω_{syn}) $\rightarrow$ rotational kinetic energy: $\frac{1}{2}J\omega^2$ with J the moment-of-inertia and ω the angular speed. As long as $P_{m,pu} > P_{e,pu}$, the machine gains kinetic energy and accelerates.

□

Equal Area Criterion: the case of a fault I

Let us consider again the system of Figure 1, where

- a **short-circuit (the fault)** occurs on one of the lines at time t and angle δ_0,
- this short-circuit is **cleared** at time $t + t_{cl}$ and angle δ_{cl} by **tripping** the line.



(a)

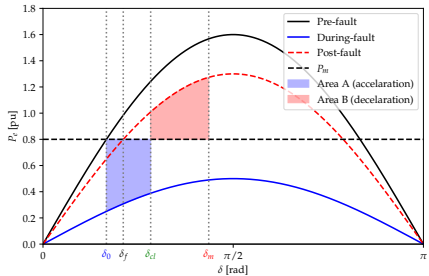
For a stable system, we highlighted that $\int_{\delta_0}^{\delta} (P_{m,pu} - P_{e,pu})\, d\theta = 0$.

Equal Area Criterion: the case of a fault II

For our short-circuit example, we can decompose the condition into:

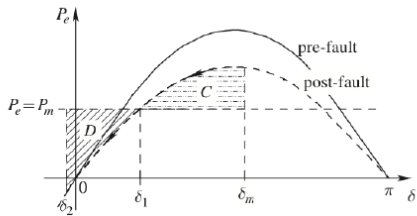
$$\underbrace{\int_{\delta_0}^{\delta_{cl}} \left(P_{m,pu} - P_{e,fault,pu}\right) d\theta}_{\text{Area A}} - \underbrace{\int_{\delta_{cl}}^{\delta_m} \left(P_{e,post-fault,pu} - P_{m,pu}\right) d\theta}_{\text{Area B}} = 0$$

- During the first part (Area A), the machine accelerates.
- After the fault (Area B), the machine decelerates and the net acceleration becomes 0.



Equal Area Criterion: the case of a fault III

At angle δ_m, there is still a mismatch between the electrical power and the mechanical power.
Thus the machine swings back, from δ_m to δ_2.

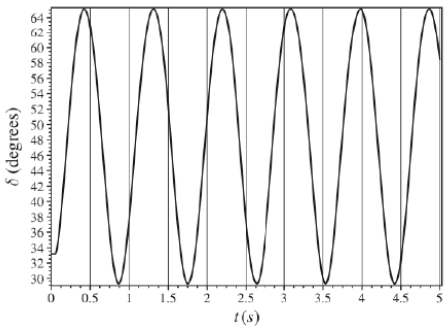


Exercises

- Compute δ_m, the angle reached just after clearing the fault.
- Compute δ_2, the angle reached when swinging back.

Equal Area Criterion: the case of a fault IV

Without any damping (kinetic energy losses), the system oscillates indefinitely between angle δ_2 and δ_m.



However, in a **real system**, the **damping** would cause the machine to settle down at an angle δ_1 (new equilibrium point).

Reminder: Synchronous machines have *damper windings*. In perfect steady state, the magnetic fields produced by both the stator and the rotor are fixed relative to the rotor so there is no current in the dampers. On the other hand, if the rotor moves with respect to the magnetic field, the current induced in the dampers create a damping torque according to Lenz's law.

Critical Clearing Angle

Let us come back to our definition of stability for our little system:

$$\underbrace{\int_{\delta_0}^{\delta_{\mathbf{cl}}} \left(P_{m,pu} - P_{e,fault,pu}\right) d\theta}_{\text{Area A}} - \underbrace{\int_{\delta_{\mathbf{cl}}}^{\delta_m} \left(P_{e,post-fault,pu} - P_{m,pu}\right) d\theta}_{\text{Area B}} = 0$$

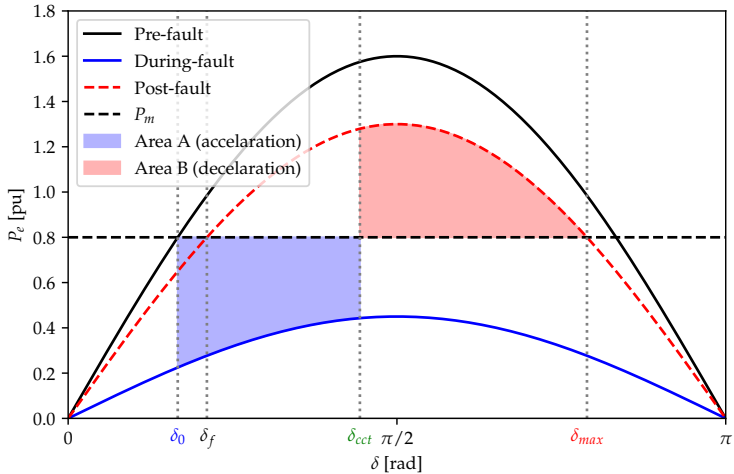
What are the conditions to ensure there exists $\delta_{\mathbf{cl}}$ such that Area A = Area B?

We introduce the concept of critical **clearing angle** δ_{cct}. Past this point, Area A will always be greater than Area B, such that the system cannot stabilize. We associate the critical clearing time *CCT* to the critical clearing angle, *i.e.* the maximum time allowed to clear the fault before the system gets unstable.

Critical Clearing Angle: Mathematical formulation

$$\underbrace{\int_{\delta_0}^{\delta_{cct}} \left(P_{m,pu} - P_{e,fault,pu}\right) d\theta}_{\text{Area A}} - \underbrace{\int_{\delta_{cct}}^{\delta_{max}} \left(P_{e,post-fault,pu} - P_{m,pu}\right) d\theta}_{\text{Area B}} = 0 \tag{3}$$

Critical Clearing Angle: Visual representation



Critical Clearing Angle: Solution procedure

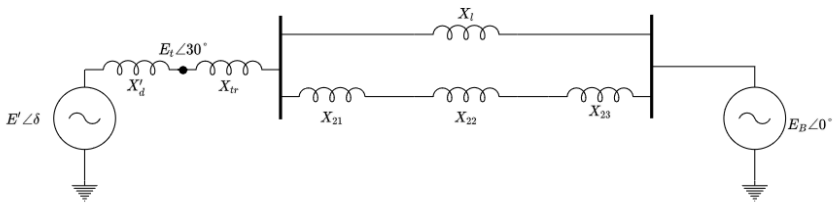
Solution procedure for Critical Clearing Angle and Time determination

1. Determine
   - the angle δ_0 such that $P_{e,pre-fault} = P_m$
   - the angle δ_{max} such that $P_{e,post-fault} = P_m$.
2. Then solve equation (3) to get δ_{cct}.
3. Finally, using the **swing equation**, one can determine the critical clearing time *CCT*, *i.e.* the time needed to reach the angle δ_{cct}.

Practical Example

Problem statement

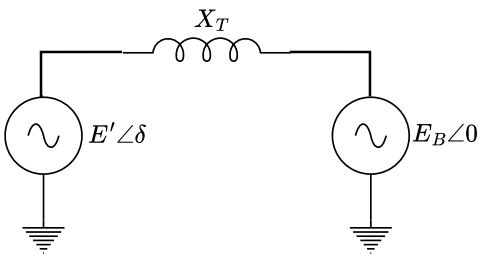
Let us consider the following system:



$$X'_d = 0.3, X_{tr} = 0.5, X_l = 1, X_{21} = 0.5, X_{22} = \tfrac{1}{6}, X_{23} = \tfrac{1}{3}$$
$$\bar{E}_t = 1e^{j\pi/6}, \bar{E}_B = 1e^{j0}.$$

Equivalent

We can derive the following equivalent:



where

$$X_T = X_{tr} + X'_d + \frac{X_l\,(X_{21} + X_{22} + X_{23})}{X_{21} + X_{22} + X_{23} + X_l} = 1.3pu.$$

The power transfer from the machine to the infinite-bus system takes the following form:

$$P = \frac{E'E_B}{X_T}\sin\delta$$

Find $\bar{E}'$, P_e and P_m I

1. Current from $\bar{E}_t$ to $\bar{E}_B$

$$\bar{I}_{t\to B} = \frac{\bar{E}_t - \bar{E}_B}{jX_T - jX'_d}$$

2. $\bar{I}_{t\to B}$ is the same as the one from $\bar{E}'$ to $\bar{E}_t$

$$\bar{E}' = jX'_d\bar{I}_{t\to B} + \bar{E}_t = 1.05095e^{j38.2057/180}$$

3. Find P_e

$$P_e = \frac{1.05095\ 1\ \sin(38.2057/180)}{1.3} = 0.5$$

Find $\bar{E}'$, P_e and P_m II

4. Initially, the system is at equilibrium.

$$P_m = P_e = 0.5$$

In the following, we will compute the maximum power outputs for three different conditions: Pre-fault, During-Fault and Post-Fault. It will allow us to determine the 3 different $P - \delta$ curves.

Pre-fault condition

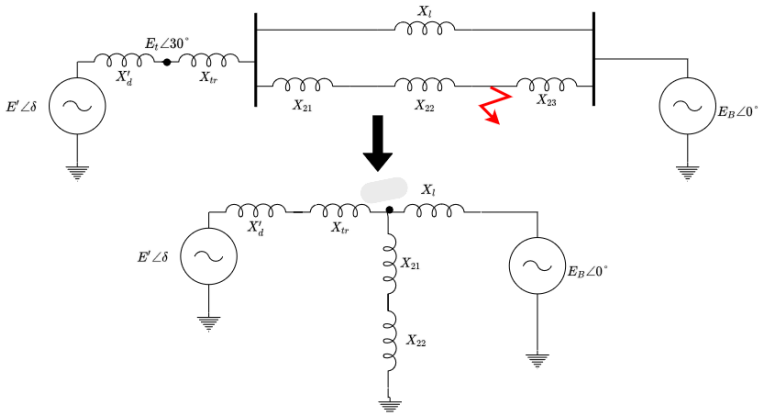
We compute the maximum power output while assuming E' does not change. *If the transient stability study lasts a second or less, it is reasonable to consider, as a first-order approximation, that the exciter of the synchronous machine cannot respond in such short amount of time. Hence, E' does not change*

Maximum power output:

$$\hat{P}_{e,bf} = \frac{E'E_B}{X_T} = 0.808$$

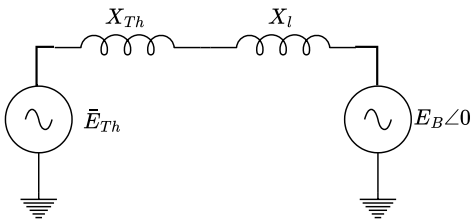
During-fault condition

A short-circuit occurs between lines 22 and 23. The circuit topology changes.



During-fault condition: Thévenin equivalent

We can derive the following Thevenin's equivalent



where

$$\bar{E}_{Th} = \bar{E}' \frac{X_{21} + X_{22}}{X_d + X_{tr} + X_{21} + X_{22}} = 0.478 e^{j38.206/180}$$

and

$$X_{Th} = \frac{1}{\frac{1}{X'_d + X_{tr}} + \frac{1}{X_{21} + X_{22}}} = 0.364$$

and then get the maximum power output:

$$\hat{P}_{e,df} = \frac{E_{Th} E_B}{X_{Th} + X_l} = 0.35$$

Post-fault condition

The line 2 is tripped to clear the short-circuit (a switch is open between the bus bar and X_{21}).

The impedance of the path connecting the machine to the infinite-bus system becomes $X'_d + X_{tr} + X_l = 1.8$. The maximum power output is:

$$\hat{P}_{e,pf} = \frac{E'E_B}{X'_d + X_{tr} + X_l} = 0.584$$

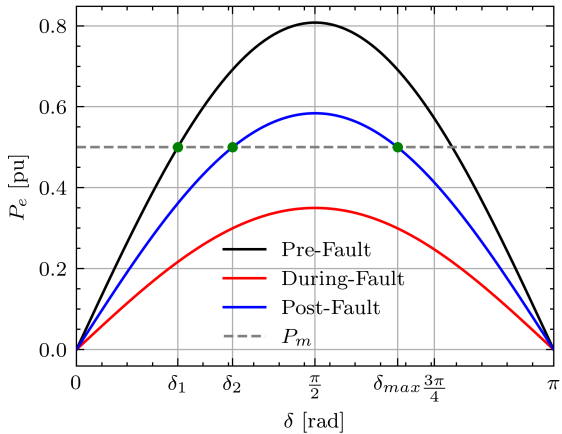
Practical Example

The final $P - \delta$ curves are shown here under, with

$$\delta_1 = \arcsin\left(\frac{P_m}{\hat{P}_{e,bf}}\right) = 0.667$$

$$\delta_2 = \arcsin\left(\frac{P_m}{\hat{P}_{e,pf}}\right) = 1.028$$

$$\delta_{max} = \pi - \delta_2 = 2.114$$



Critical clearing angle δ_{cct} I

$$\underbrace{\int_{\delta_0}^{\delta_{cct}} \left(P_{m,pu} - P_{e,fault,pu}\right) d\theta}_{\text{Area A}} = \underbrace{\int_{\delta_{cct}}^{\delta_{max}} \left(P_{e,post-fault,pu} - P_{m,pu}\right) d\theta}_{\text{Area B}}$$

Area A

$$\begin{aligned}\int_{\delta_0}^{\delta_{cct}} \left(P_{m,pu} - P_{e,fault,pu}\right) d\theta &= P_{m,pu}\left(\delta_{cct} - \delta_0\right) - \int_{\delta_0}^{\delta_{cct}} \hat{P}_{e,df} \sin\theta d\theta \\ &= P_{m,pu}\left(\delta_{cct} - \delta_0\right) + \hat{P}_{e,df}\left(\cos\delta_{cct} - \cos\delta_0\right)\end{aligned}$$

Critical clearing angle δ_{cct} II

Area B

$$\int_{\delta_{cct}}^{\delta_{max}} \left(P_{e,pf,pu} - P_{m,pu}\right) d\theta = P_{m,pu}\left(\delta_{cct} - \delta_{max}\right) - \int_{\delta_{cct}}^{\delta_{max}} \hat{P}_{e,pf} \sin\theta d\theta$$
$$= P_{m,pu}\left(\delta_{cct} - \delta_{max}\right) + \hat{P}_{e,pf}\left(\cos\delta_{cct} - \cos\delta_{max}\right)$$

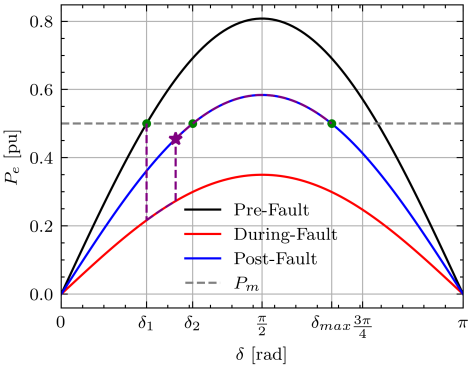
Equal-area criterion

$$\text{Area A} = \text{Area B}$$
$$P_{m,pu}\left(\delta_{max} - \delta_0\right) + \hat{P}_{e,df}\left(\cos\delta_{cct} - \cos\delta_0\right) = \hat{P}_{e,pf}\left(\cos\delta_{cct} - \cos\delta_{max}\right)$$

Critical clearing angle δ_{cct} III

We can find out the critical clearing angle.

$$\cos \delta_{cct} = \frac{P_{m,pu} \left(\delta_0 - \delta_{max} \right) + \hat{P}_{e,df} \cos \delta_0 - \hat{P}_{e,pf} \cos \delta_{max}}{\hat{P}_{e,df} - \hat{P}_{e,pf}}$$



The purple star corresponds to $\delta_{cct} = 0.8938$, and the dashed purple line shows the evolution of the angle δ. Even without damping, the system stabilizes at $\delta = \delta_{max}$ since the kinetic energy reaches 0 at that point ($\Delta\omega^2 = 0$)

Estimation of the critical clearing time *CCT* I

In order to find the critical clearing time, one would need to solve the swing equation:

$$\left(\frac{2H}{\omega_{syn}}\right)\frac{d^2\delta(t)}{dt^2} = P_{m,pu} - P_{e,pu}(\delta(t))$$

where $P_{e,pu}(\delta(t))$ is a non-linear function depending on δ; $P_{e,pu}(\delta(t)) = K(t)\sin(\delta(t))$, and with K changing non-continuously with time. We would need to solve a non-linear differential equation, which is not an easy task! Thus, let us assume that the acceleration $\frac{d^2\delta(t)}{dt^2}$ is constant over time. We have:

$$\delta_{cct} = \delta_1 + \Delta\omega_0\, CCT + a\frac{CCT^2}{2}$$

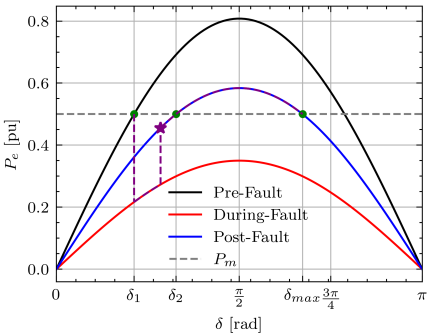
Estimation of the critical clearing time *CCT* II

where a is the constant acceleration and $\Delta\omega_0$ the initial speed assumed to be 0.

Let us assume two cases:

1. We pick the maximum acceleration for angles in $[\delta_1, \delta_{cct}]$.
2. We pick the minimum acceleration for angles in $[\delta_1, \delta_{cct}]$.

From the following figure



Estimation of the critical clearing time *CCT* III

it is clear that the acceleration is maximum for $\delta = \delta_1$, and minimum for $\delta = \delta_{cct}$ for $\delta \in [\delta_1, \delta_{cct}]$ (larger power mismatch when $\delta = \delta_1$ and smaller for $\delta = \delta_{cct}$). We have:

$$a_{min} = \frac{\omega_{syn}}{2H}\left(P_{m,pu} - \hat{P}_{e,df}\sin\delta_{cct}\right)$$

$$a_{max} = \frac{\omega_{syn}}{2H}\left(P_{m,pu} - \hat{P}_{e,df}\sin\delta_1\right)$$

With $H = 4.5$s, we can derive a lower and an upper bound for the critical clearing time *CCT*.

$$CCT_{max} = \sqrt{\frac{2\left(\delta_{cct} - \delta_1\right)}{a_{min}}} = \sqrt{\frac{2\left(0.8938 - 0.667\right)}{7.93}} = 239\text{ms}$$

Estimation of the critical clearing time *CCT* IV

$$CCT_{min} = \sqrt{\frac{2\left(\delta_{cct} - \delta_1\right)}{a_{min}}} = \sqrt{\frac{2\left(0.8938 - 0.667\right)}{9.89}} = 214\text{ms}$$

If the actual clearing time is denoted t^*, we can conclude that:

1. If $t^* > CCT_{max}$, the system is unsafe!
2. If $t^* < CCT_{min}$, the system is safe!
3. If $CCT_{min} < t^* < CCT_{max}$, we have no guarantee if the system is safe or not.

Short-Circuit Capacity and Rotor Angle Stability

Grid Strength: Subtransient Reactance and Thévenin Impedance I

- **Subtransient behavior of synchronous machines**: Immediately after a short circuit occurs ($t \approx$ 10–40 ms), magnetic flux linkages trapped inside rotor windings cannot change discontinuously. Synchronous machines behave as ideal AC voltage sources behind their **subtransient reactance** X''_d, feeding large currents (5 to 7 $\times$ I_{nominal}) into the network.

Grid Strength: Subtransient Reactance and Thévenin Impedance II

- **Thévenin equivalent and IEC 60909 standard**: Seen from any network bus i, the interconnected transmission grid can be represented by its Thévenin equivalent impedance $Z_{th,i} \approx jX_{th,i}$ and nominal voltage U_n. The **initial symmetrical short-circuit current** I_k'' and **short-circuit apparent power** S_{sc}'' are:

$$I_k'' = \frac{c \cdot U_n}{\sqrt{3}|Z_{th}|} \approx \frac{c \cdot U_n}{\sqrt{3}X_{th}}, \qquad S_{sc}'' = \sqrt{3}U_n I_k'' \approx \frac{c \cdot U_n^2}{X_{th}}$$

  where $c = 1.10$ is the voltage factor for high-voltage networks.

- **Physical meaning of Grid Strength**:

Grid Strength: Subtransient Reactance and Thévenin Impedance III

- **High** I_k'' **/ High** S_{sc}'' $\iff$ **Low** X_{th}: **Strong grid** with stiff voltage support and tight electrical coupling.
- **Low** I_k'' **/ Low** S_{sc}'' $\iff$ **High** X_{th}: **Weak grid** with high electrical impedance to surrounding sources.

Impact of Short-Circuit Level on the Equal Area Criterion I

Recall the electrical power delivered by a synchronous machine to an infinite bus:

$$P_e(\delta) = \hat{P}_e \sin\delta = \frac{E'V}{X'_d + X_{th}} \sin\delta$$

Post-fault power-angle curve:

- The peak post-fault power $\hat{P}_{e,pf} = \frac{E'V}{X'_d + X_{th,pf}}$ depends directly on the grid Thévenin reactance $X_{th,pf}$.
- Higher short-circuit capacity $S''_{sc} \implies$ smaller $X_{th,pf} \implies$ **much higher** $\hat{P}_{e,pf}$.

Impact of Short-Circuit Level on the Equal Area Criterion II

Equal Area Impact:

- **Decelerating Area (Area B)** expands:

$$\text{Area B} = \int_{\delta_{cct}}^{\delta_{max}} (\hat{P}_{e,pf} \sin\theta - P_m) d\theta$$

- $\delta_{max} = \pi - \arcsin(P_m/\hat{P}_{e,pf})$ shifts to the right, widening the permissible angular swing.

Critical Clearing Time:

- With a larger decelerating capacity, the system can tolerate a larger accelerating area (Area A).
- The critical clearing angle δ_{cct} increases.
- The **Critical Clearing Time** (*CCT*) **increases**, giving protection relays more time to clear faults safely.

Small-Signal Synchronizing Torque Coefficient K_s

Linearizing the swing equation around an operating angle δ_0 ($M\frac{d^2\Delta\delta}{dt^2} + K_s\Delta\delta = 0$):

$$K_s = \left.\frac{\partial P_e}{\partial \delta}\right|_{\delta_0} = \frac{E'V}{X'_d + X_{th}} \cos\delta_0$$

- A higher short-circuit capacity (X_{th} small) directly increases K_s.
- The pre-fault operating angle $\delta_0 = \arcsin\left(\frac{P_m(X'_d+X_{th})}{E'V}\right)$ is smaller $\implies \cos\delta_0 \approx 1$.
- **Physical consequence**: A stiff electromechanical coupling ("stiffer magnetic spring") prevents monotonic loss of synchronism and increases the oscillation frequency:

$$\omega_n = \sqrt{\frac{K_s}{M}}$$

Spatial Propagation of Fault Voltage Dips

Short-circuit strength also governs the spatial severity of voltage dips during faults:

- **Strong grid (high short-circuit current I_k'')**:
  - Fault-induced voltage depressions remain localized close to the fault impedance.
  - Surrounding generators maintain healthy terminal voltages and continue delivering electrical counter-torque P_e.
- **Weak grid (low short-circuit current I_k'')**:
  - The voltage collapses across vast electrical distances ($V \to 0$).
  - Distant generators lose their electrical counter-torque simultaneously ($P_e \to 0$), causing wide-area multi-machine rotor acceleration.

Energy Transition: Inverter-Based Resources vs. Synchronous Fleets I

Inverter Fault Current Limitation:

- Synchronous generators naturally inject 5–7 $\times$ I_n during faults due to physical subtransient rotor flux conservation.
- Inverters (PV, Wind) are strictly limited by semiconductor thermal limits to 1.1–1.2 $\times$ I_n.

System Stability Impact:

- Decommissioning synchronous plants causes local short-circuit capacity S''_{sc} to collapse ($X_{th} \uparrow$).
- Remaining synchronous units see a weaker grid: their peak power $\hat{P}_e$ drops, decelerating Area B shrinks, and *CCT* shortens.

Energy Transition: Inverter-Based Resources vs. Synchronous Fleets II

Modern Mitigation Strategies in Low Short-Circuit Grids:

- **Synchronous Condensers**: De-clutched synchronous machines (no prime mover) providing pure subtransient fault current (I''_k), physical inertia (H), and reactive voltage support.
- **Grid-Forming (GFM) Inverters**: Advanced control schemes that operate converters as instantaneous voltage sources behind a virtual impedance, delivering instantaneous power during frequency and voltage disturbances.

Extension on dynamics

Swing equations with damping coefficient

$$\frac{d\Delta\omega(t)}{dt} = \frac{1}{2H}\left(P_m - P_{e,max}(t)\sin(\delta(t)) - D\Delta\omega(t)\right)$$

$$\frac{d\delta(t)}{dt} = \omega_0\Delta\omega(t)$$

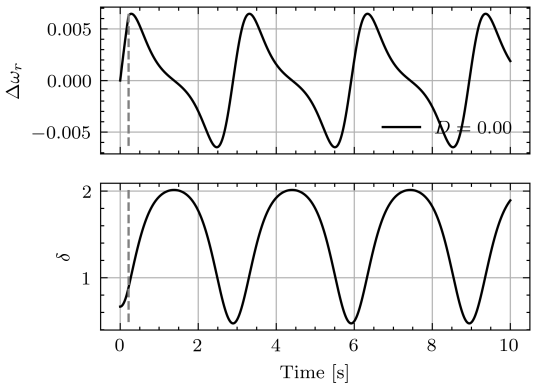
Euler discretization

$$\Delta\omega_{t+1} = (1 - \tau\frac{D}{2H})\Delta\omega_t + \frac{\tau}{2H}\left(P_m - P_{e,max}^{k\tau}\sin(\delta_t)\right)$$

$$\delta_{t+1} = \delta_t + \tau\omega_0\Delta\omega_t$$

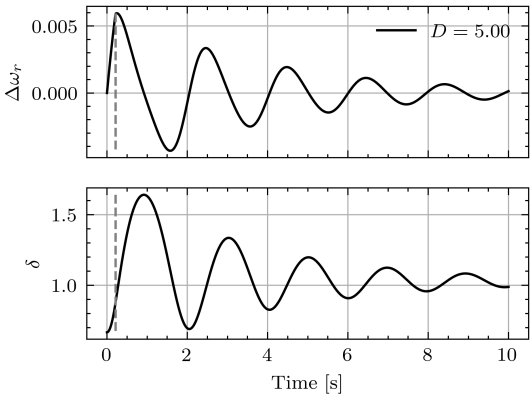
where τ is the time step and where we explicitly denote that $P_{e,max}^{k\tau}$ is changing over time in a discrete fashion (pre-fault$\rightarrow$during-fault$\rightarrow$post-fault) through the superscript $k\tau$.

Results **without** damping, $t_{cl} = 0.214s \leq CCT$



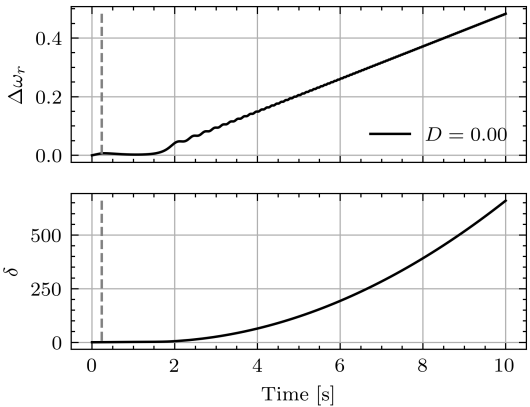
The fault is cleared after 214 ms (lower bound on the clearing time). As expected from previous calculations, the system is safe.

Results **with** damping, $t_{cl} = 0.214 \leq CCT$



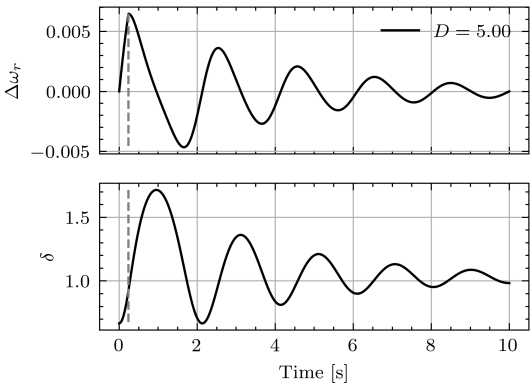
The fault is cleared after 214 ms (lower bound on the clearing time). As expected from previous calculations, the system is safe. The damping adds energy dissipation, which allows the system to stabilize around $\delta_2 = 1.028$

Results **without** damping, $t_{cl} = 0.239s \geq CCT$



The fault is cleared after 239 ms (upper bound on the clearing time). The machine loses synchronism.

Results **with** damping, $t_{cl} = 0.239s \geq CCT$



The fault is cleared after 239 ms (upper bound on the clearing time). The system is stable thanks to the damping effect (energy dissipation).

Dynamic Security Assessment

Dynamic Security Assessment goal

Main question that Dynamic Security Assessment (DSA) should answer:

Imagine a set of major, yet credible contingencies, can the system resist such events without jeopardizing its integrity?

If yes, then the states are determined secure $\rightarrow$ the system trajectories do not bring the states inside an unsafe set.

Dynamic Security Assessment principle

- Based on fast time domain contingency simulations.
- Study main stability issues such that: Voltage, Transient and Small-Signal (not covered throughout this course).
- Start from actual and future operating points.
- Has to be visual for the operators $\rightarrow$ show the type of instability, where it comes from and even possible solutions.

Main types of DSA analysis

1. Off-line analysis

- They are subjected to forecast errors $\rightarrow$ system security cannot be taken for granted.
- But they can be performed with no constraints on time (performed day-ahead in order to set up a recommended operating schedule)

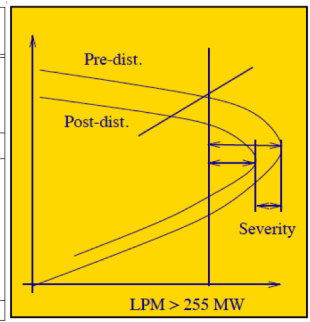
2. On-line analysis

- Not prone to forecast errors since they are based on real-time information.
- But need to be fast.

Voltage stability assessment

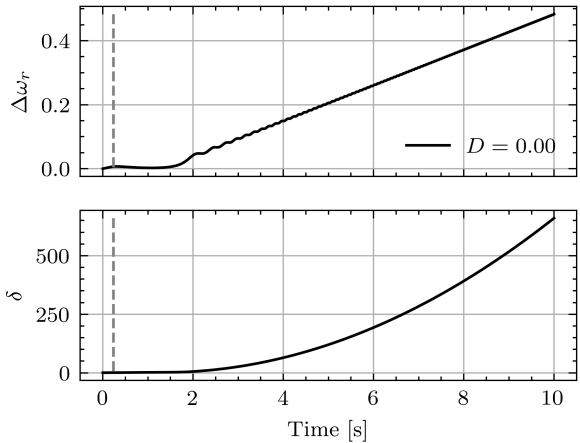
- Mainly static analyses.
- Goal is to ensure a sufficient load power margin.
- List of contingencies: all single-component outages.

Numéro état	4		
Num.Variante	6		
Consignations	(CHA61COR)		
Charge	4927.MW+1502.Mvar		
Compensation	285		
Sites	Prod (MW)	Q (Mvar)	Rés. (Mvar)
Cordemais	548	144	237
Flamanville	0	0	0
Chinon	1840	540	794
Cheviré	0	125	414
TACs	0	0	0
St. Laurent	?	?	?
Blayais	2500	650	1558
Marge de P.C.	1002.MW+308.0Mvar		



Transient stability assessment

We want to avoid that:



- Simulations with different three-phase short-circuits at various locations.
- And with different clearing schemes!
- Very time consuming computations.

Trends

Nowadays

- Integration of large off-shore wind parks, and PV systems without reinforcing the network.
- Large power plants are decommissioned, and conventional generators are no longer accepted in urban areas.
- New units are built far away from load centers.
- Hard to build new lines because of public debates and environmental constraints.

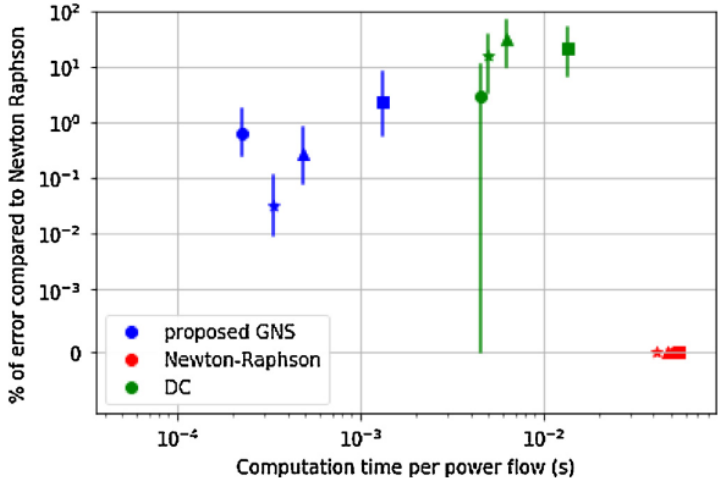
This leads to a weaker electrical system, more prone to stability issues. There's a need for tools that can quickly perform security assessment to guarantee a secure system.

Example 1: static analysis I

- For static analyses, we rely on power flow solvers to estimate the state of the system.
- With increasing penetration of RES, probabilistic approaches are envisioned to perform risk assessment.
- But traditional power flow solvers based on Newton-Raphson methods are too slow.
- Usage of AI tools to derive approximate solutions of the power flow equations.

Donon, B., Clément, R., Donnot, B., Marot, A., Guyon, I., & Schoenauer, M. (2020). Neural networks for power flow: Graph neural solver. Electric Power Systems Research, 189, 106547.

Example 1: static analysis II

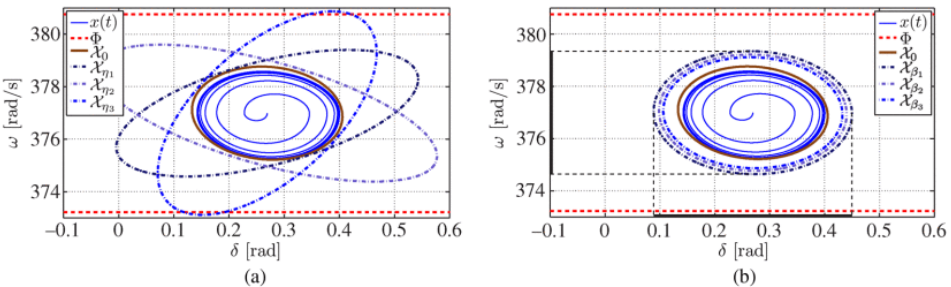


Example 2: dynamic analysis I

- For dynamic analyses, we study the time-evolution of a power system trajectory (e.g. internal angle in transient stability)
- It requires solving the differential-algebraic equations for multiple scenarios (different short-circuit locations, different clearing schemes).
- Usage of reachability analysis techniques: gives the reach set, i.e., the set that contains all possible system trajectories.

Chen, Y. C., & Dominguez-Garcia, A. D. (2012). A method to study the effect of renewable resource variability on power system dynamics. IEEE Transactions on Power Systems, 27(4), 1978-1989.

Example 2: dynamic analysis II



References I



N. Hatziargyriou, J. Milanovic, C. Rahmann, V. Ajjarapu, C. Canizares, I. Erlich, D. Hill, I. Hiskens, I. Kamwa, B. Pal, *et al.*, "Definition and classification of power system stability–revisited & extended," *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3271–3281, 2020.